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Describe the pathway taken by an oxygen molecule from an alveolus to the blood.

**[2 marks]**

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Explain how **one** feature of an alveolus allows efficient gas exchange to occur.**[2 marks]**

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**Question 9 continues on the next page**

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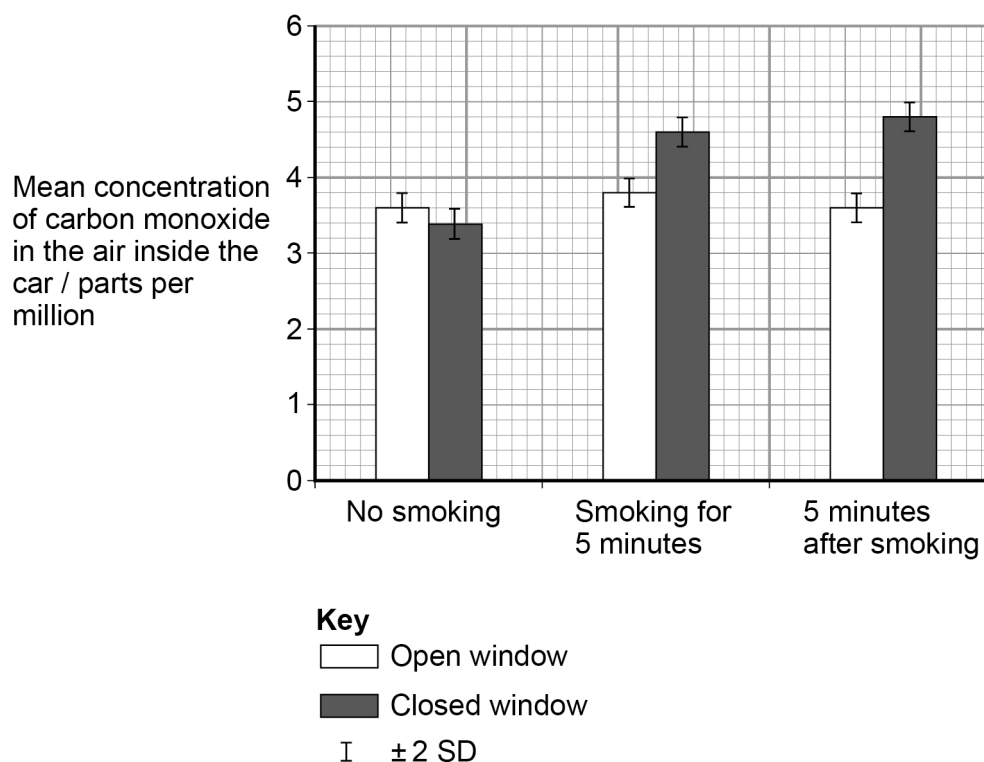
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Carbon monoxide is a poisonous gas that is present in cigarette smoke. This carbon monoxide can be absorbed into the blood where it binds with haemoglobin.

Scientists investigated the concentration of carbon monoxide in cars in which people were smoking or not smoking. They measured the concentration with the car windows open and closed. **Figure 7** shows the scientists' results as they presented them. A value of  $\pm 2$  standard deviations from the mean includes over 95% of the data.

**Figure 7**



In England, in October 2015, a law was introduced making it illegal to smoke in a car carrying someone who is under the age of 18.

'It is dangerous to smoke when a child is in the car. Higher levels of deadly toxins can build up, even on short journeys, and children breathe faster than adults, meaning they inhale more of the deadly toxins.'

Use the information provided **and** the data in **Figure 7** to evaluate the politician's statements.

[illegible]

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